

Computer Vision Practical Report:

DISPLAY VIDEO IN FAST & SLOW MOTION

1. AIM

To read an input video using OpenCV and display the video in both **fast-motion** and **slow-motion** by controlling the playback speed.

2. PROCEDURE

- Load the input video using `cv2.VideoCapture()`.
- Read the video frames using `cap.read()`.
- Display the frames using `cv2.imshow()`.
- For **slow motion**, increase the delay between frames using `cv2.waitKey()`.
- For **fast motion**, decrease the delay between frames.
- Press the 'q' key to stop the video playback.
- Release the video using `cap.release()`.
- Close all display windows using `cv2.destroyAllWindows()`.

3. PROGRAM CODE

```
import cv2
import matplotlib.pyplot as plt

video_path = 'E7-VID.mp4'

cap = cv2.VideoCapture(video_path)
frames = []

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    frames.append(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
```

```
cap.release()

if len(frames) > 0:
    print("1 - Fast Motion")
    print("2 - Slow Motion")
    choice = input("Enter your choice (1 or 2): ")

    step = 5 if choice == "1" else 1
    pause_time = 0.001 if choice == "1" else 0.1

    plt.ion()
    fig, ax = plt.subplots(figsize=(8, 5))
    im = ax.imshow(frames[0])
    ax.axis('off')
    ax.set_title("Fast Motion" if choice == "1" else "Slow Motion")

    for i in range(0, len(frames), step):
        im.set_data(frames[i])
        fig.canvas.draw_idle()
        plt.pause(pause_time)

    plt.ioff()
    plt.show(block=True)
```

4.OUTPUT

Slow Motion



5. RESULT

The Python script executed successfully. The input image 'E7-IMG.png' was loaded.